Research Expectations and Background Information

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Introduction

This document contains my expectations of myself and those working in my research group. I also include a (highly incomplete) list of resources and background topics which I find are critical to my work. As with anything of this nature, this document and my opinions within are highly biased to my own career thus far and my experiences. The intention of this document is not to add to the stress and anxiety of research life, but to give clear expectations to both the mentor and mentee to position everyone for maximal success.

Expectations for Research

This section presents my expectations for both the mentor and mentee positions in my research group. This is an incomplete list, modeled in part after a similar list from my postdoc mentor Joel Leja (and, by transitivity, his former postdoc mentor Charlie Conroy), but with several additions from my own experiences. Suggestions are always welcome.

Responsibilities and expectations of the mentor

1. To support the development of the mentee's career path. This includes anything related to research, academics, professional development, etc., and broader mentorship on career/life choices, work-life balance, or other related topics.
2. To offer guidance to the mentee on research projects designed to develop technical skills and have broader impacts on the remainder of their career.
3. To assist the mentee in developing connections with more senior scientists who may serve as references and additional sources of mentorship and collaboration.
4. To ensure a welcoming and inclusive environment designed to encourage engagement and a sense of belonging for everyone in the group. This includes the development of an environment in which all participants are comfortable discussing diverse scientific topics, as well as respectfully challenging others and being challenged themselves.
5. To recognize the strengths and weaknesses of the mentee, and adapt their mentoring strategy to address them appropriately.
6. To acknowledge their own biases and shortcomings in all facets, but particularly in the mentor/mentee relationship, and work toward rectifying them.
7. To provide constructive and timely feedback, both on the short term (week/month) and long term research and career progress of the mentee.
8. To be available as a confidant in the case of bullying, harassment, or any other difficult professional or social issue. In conjunction with (3), this includes ensuring that the mentee has external avenues for the mediation of issues that cannot be directly discussed with the mentor.
9. To maintain flexibility for the evolving needs of the mentee over time.

Responsibilities and expectations of the mentee

1. To take ownership of their research. It is ultimately the first author of any given research product (papers, presentations, etc.) who both gets the recognition and carries the responsibility of communicating, modifying, and defending the work.
2. To understand the broader impacts of their work. This includes both how their work immediately impacts their subfield, but the understanding of astronomy (or even science as a whole).
3. To grow their skills as a researcher. It is expected (at any level) that one will improve their skills with time, and with that comes an increase in the expectations for self-reliance and agency over their research.
4. To be ambitious. A commonality that I (Nikko) have witnessed is that many very-early-career researchers (particularly in their first few years of graduate school) feel that they are 'too young/early' to lead projects from their own ideas, write PI proposals, etc.. Taking on more ambitious projects early on can be

incredible learning experiences (even if proposals get rejected), and only set up for more success later on.

5. To measure productivity on reasonable timescales. Everyone has ‘slumps’ where it feels as if they are not accomplishing anything for weeks at a time, but issues holding back progress for a month or more should be discussed and the research plan should be reassessed to develop more effective work habits.
6. To be mindful of themselves and their work-life balance. It is unavoidable that there will be times when other parts of life are more demanding of one’s time, and it is entirely okay to be fluid and reprioritize as needs arise.
7. To be mindful in their scientific communication. This refers to all facets of the research process, including presentations, papers, proposals. This is discussed in more detail later in this document.
8. To be proactive in their networking. Astronomy is a small field, and there is a potential future collaborator/colleague/employer at any given talk or conference. Though it may not seem to be the case, it does not take long to know many/most of the people in your subfield and beyond in such a small field.
9. To attend talks, at least one per week. This is a highly effective way to broaden one’s scope of scientific knowledge beyond just their immediate collaboration/subfield. Additionally, the presence of external speakers is a low-effort means of networking.
10. To ask questions, especially ‘stupid’ ones. This is how one strengthens their fundamentals and expands their knowledge, while patching holes in their scientific foundation (which everyone has, regardless of level).
11. To read literature. This includes both foundational papers in one’s field, as well as new papers to keep up with the evolving forefront of research. Additionally, reading the literature from other fields is a key way to expand one’s peripheral knowledge. This is discussed in more detail later in this document.
12. To practice professional development. This includes the commonly discussed topics of CVs, personal branding, etc., but also includes learning new technical skills to indirectly improve their research toolbox (e.g., learning new programming tools/methods).
13. To contribute constructively to the equitable and inclusive research environment.

Responsibilities and expectations of both the mentor and mentee

1. To self-advocate and communicate their needs and expectations. It is unavoidable that there will be times in any mentor/mentee relationship that the needs or expectations of either party may be unsatisfied, and as such it is critical that those needs are communicated so that they can be assessed and adapted to accordingly.

2. To meet with a regular cadence appropriate for the evolving needs and expectations of the mentee and the project. This is most often going to be ~weekly, though in higher-output periods (e.g., proposal season, a paper nearing submission, etc.) this could be more frequent.
3. To come prepared to meetings. For the mentee, this means being ready to discuss what the mentee has done/is doing, what they will do, and what issues they are having or assistance they need. For the mentor, this means being ready to provide feedback and guidance on next steps.

Research Introduction

I generally study the evolution of galaxies through cosmic time, stretching all the way to the formation of the very first galaxies over 13 billion years ago. To do so, I study spectra from space-based observatories, particularly the *James Webb Space Telescope* (*JWST*). *JWST* has discovered the most distant galaxies known today and has completely revolutionized the field of galaxy evolution and astronomy as a whole.

Specifically, I study the conditions of stars and black holes at early cosmic times. This includes the formation of the very first stars (so-called Population III) from primordial gas, to their deaths and subsequent enrichment of the interstellar medium (ISM) with heavier elements to then form the next generations of stars. At the same time, the black holes in the centers of galaxies were forming and growing to become the supermassive black holes we observe throughout cosmic time. My ultimate research goals are to answer the questions:

How and when did the first stars and galaxies form?

How and when did the first black holes form?

How did both the stars and black holes evolve over cosmic time?

Astronomy and Physics Background

A general background in many facets of astronomy is critical to being a productive researcher in any subfield. There are many resources for introductory material, including the classic Big Orange Book (BOB): *An Introduction to Modern Astrophysics* by Carroll and Ostlie (this was my very first astronomy textbook, which I still have and occasionally refer to today). Functionally, background from texts like this or from course work in astronomy, physics, chemistry, and math/statistics is enough to begin doing astronomy research. Doing research is going to teach more in the long run than any

other form of learning, but it's often unavoidable that it will be more specific and directed in scope.

Here I list a number of background questions to consider that are central to my work (in no particular order):

- What is a spectrum? What information can we gain from emission/absorption lines?
- What do spectra from different types of objects look like?
- What do spectra at different wavelengths look like?
- What is dust and how does it affect our observations?
- How do stars form? How do stars die?
- What is the interstellar medium?
- How do black holes form and grow?
- How do black holes interact with their host galaxies? What is an 'active galactic nucleus' (AGN)?
- How does the Universe and the stuff within it change as a function of cosmic time?
- What biases do our observations suffer? How do we deal with them in the interpretations of our results?

Specific Background and Resources

For a more technical background/next step into reading papers, I have created a SciX library of some major papers in extragalactic astronomy. I also recommend *Astrophysics of Gaseous Nebulae and Active Galactic Nuclei* by Osterbrock and Ferland for a great background on the physics of the interstellar medium and active galactic nuclei as observed through spectroscopy. You can also find many great review papers from the Annual Reviews of Astronomy & Astrophysics (ARA&A) by querying ADS, most of which have introductions suitable for any level of expertise. These are great resources and there should be some relevant review for almost any topic, including the very first astronomy paper I ever read. More on reading papers below.

I certainly don't require that anyone has an all-encompassing and hole-free background to begin doing research (nobody knows everything, regardless of level). However, having a stronger background will almost always allow for faster improvement and more effective research.

Science Communication

Research is only useful if properly communicated. I strongly believe that the accessibility and effective communication of results is one of the most important skills to learn as a researcher, regardless of field. The most effective ways to learn how to communicate science is to read and listen to others and to present science yourself.

This means: reading papers, going to talks, giving talks, writing papers (and proposals, etc.). All of these are critical to becoming an effective science communicator.

Papers

Before you can read a paper, you have to know where to find papers:

- Science Explorer (SciX) is probably the single most important resource for finding papers (or other materials). You can search by author/year/title/keyword etc., and find the most complete listing of publications from much of the last century.
- arXiv is a popular repository of physics/math/astronomy/etc. papers which updates daily. Anyone can submit to arXiv regardless of journal publication status.
- benty-fields is a digest of the arXiv which uses machine learning to rank papers by their relevance to you based on your previous reading. Many groups use benty-fields to host journal clubs.
- astrobites/astrobitos (ES): a reader's digest of astro-ph articles written by graduate students. Focuses deeply on removing technical jargon as a barrier to entry to maximize accessibility for non-experts.

General advice for reading a paper (and a paper about how to read and present papers):

- Try to read papers often; I try to scroll through benty-fields every day, but realistically do about three times per week.
- Every once in a while, try to read a paper that isn't directly in your subfield.
- Nobody is going to fully understand everything about a paper immediately. It is totally okay to read the abstract/conclusions of a paper and decide that it's not worth reading the rest.
    - Note: this is true for papers that aren't directly in the critical path of your research. Sometimes you unfortunately have to slog through a very dense or poorly communicated paper to get the information you need.

- If you read the abstract and decide to continue, start with the figures. Don't get bogged down in the text immediately. Try to find the one or two critical figures that make the authors' arguments.
- Come away with one or two key points, and then move on to the text only if you want more detail.

Talks

Go to whatever talks you can, even when they aren't on subjects you aren't deeply knowledgeable about. Often, if the speaker is an effective communicator, there will still be useful information for you. Additionally, you will learn effective versus ineffective strategies for communicating certain information, both verbally and visually through slides.

The next step is to start giving talks yourself. Often, this can start with as small of an audience as your immediate research group. Most departments/colleges/universities have some sort of student research presentations, often semesterly or yearly. These can be great practice for presenting to an audience with a wide range of expertise. I also highly recommend giving outreach (so-called 'public level') talks, which often come with questions you wouldn't normally get from subject matter experts. I personally think that questions like 'but what if X happens?' or 'why does Y exist at all?' can be the most fundamental and make you think the hardest, and often come only from more general audiences.

Always remember: you are the expert on your research. Being asked questions is basically always a good thing, even if it doesn't feel like it at the time. I guarantee that you will give some talk at some point in your career where it feels like you are getting berated by (often one specific member of) the audience; ask around - everyone has stories. Questions occur when people are interested in what you are talking about; it's usually better to have 'bad' questions than no questions at all. Often the bad/negative questions stem from misunderstanding, so do your best to clarify the situation. Additionally, it is entirely okay to say that you don't know something and move on; it's better to do this than to dig a hole of misinformation.

The key point: practice as much as you can, to whatever audience you can.

Data Visualization

Often, our scientific arguments come in the form of plots/figures, so it is imperative that we make them as digestible and accessible as possible. The field of data visualization has been hotly discussed for over a century, and I make no attempt to summarize all of its history here. However, it would be very useful to familiarize yourself with the basic principles.

First, portray your data accurately and honestly. Secondly, keep it as simple as you can while getting the point across (don't be fancy just for the sake of being fancy). Make all of your plots and figures with mindfulness of your audience. It is often very useful to have people who are not experts in your field to look at your figures, and ask them some questions:

- What do you get out of this figure?
- Did you need more information (e.g., legends, captions)?
- How long did it take you to fully digest this figure?
- Were any of the components of the figure unnecessary/distracting?

Often we find that we stare at our own work too much and forget that there are real people who will be looking at it someday, and they likely won't come to us to ask us to explain things. First impressions are important, and people likely won't take ten minutes to digest an overly complicated figure. It's not always possible to make every figure immediately understandable (we all have examples of this), but make all of your figures being mindful of the fact that someone who isn't you or your collaborators will be looking at it one day.

Side note about colors: Be mindful of color deficiencies when choosing color palettes. The easiest resolution to benefit the widest audience is to avoid red and green in one plot, but the theory behind color schemes and color deficiencies goes very deep. Use appropriate colormaps for your data (e.g., avoid using diverging colormaps for sequential data). Most of the time you are best off using a simple 'less to more' colormap (e.g., 'Reds' in matplotlib for redshift). And please, avoid rainbow colormaps in almost all situations unless you are plotting a literal rainbow. Remember the old adage: get it right in black and white.

The key point: present all of your results with accessibility in mind.

Programming

The overwhelming majority of modern astronomy research is done in python, IDL, julia, C/C++, or occasionally other languages. Python tends to be the popular choice (and my

personal strongest language), and is my recommendation for a first (or only) programming language. Several of the resources I share here are python-specific, so if you are set on using a different language you may have to search more.

For a high-level overview of programming/computer science, I recommend turning to freely-available resources on the internet (these are things I have used myself). YouTube is a phenomenal resource for almost anything you might want in terms of programming languages and specific packages/libraries, and has a lot of information about general design principles.

- Harvard CS50: probably the most famous intro to computer science course, teaches the very basics. Touches on languages including scratch, C/C++, and python.
- Complete Python Course for Beginners: There are a million of these beginner python courses on YouTube, but this is the one I remember liking the most. I recommend coding alongside the lessons.

Functionally, my research consists of some specific uses on the programming side. There will be many options other than those I present here, so feel free to explore alternatives (though what I show will tend to be the industry standard):

- Making plots: this is often the most important facet of research in observational astronomy, as we tend to present our arguments through figures.
    - Matplotlib: the premier plotting library in python. Can do everything from basic scatter plots to interactive visualizations. I use this quite literally every day.
    - Seaborn: another plotting library with some more specific use cases than matplotlib; I recommend only looking into after being comfortable with matplotlib
- Data management/data analysis: this is the backbone from which all research arises. Here are some of the most important python libraries:
    - Numpy: basically a requirement for everything in python. Does everything from basic arithmetic to array manipulation. Will be a dependency for many other packages in python.
    - pandas: a wonderful tool for working with dataframes (think spreadsheets). Most data in astronomy are tabular, so knowing how to work with data in this form is critical.
    - Scipy: for algorithms like integration/interpolation/optimization/curve fitting.
- Astronomy-specific programming:

    - Astropy: has many useful tools from reading .fits files (a very common file format in astronomy) to working with astronomical coordinates and units.

Other programming/python recommendations:

- Be mindful of things like readability and clarity of code even if you are just starting out. It's a lot easier to start with good habits than spend years ingraining bad ones.
    - The Zen of Python: 19 principles governing python design
    - PEP 8 style guide: formatting and other recommendations for writing python code
    - Learn how to write functions and/or classes as soon as you feel comfortable with the basics. These improve readability and functionality of your code *significantly*, and any professional employer will expect you to be able to use them.
- Familiarize yourself with the Python Standard Library. Chances are there are tools which already exist for most of the basic tasks you need to perform.
- Pick an integrated development environment (IDE) or other code editor that works for you. I use Visual Studio Code, but many use Spyder, Jupyter, or others.
- Make an account on GitHub and start to learn the basics of git. This is the modern standard for version control.